Seat No.	

Set P

B.Sc. (E.C.S.) (Semester - I) (NEP CBCS) Examination: March/April – 2026
Digital Electronics and Microprocessors (G06-0103)

Day & Date: Wednesday, 29-04-2026 Max. Marks: 30
Time: 12:00 PM To 01:30 PM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose the correct alternative. **06**

1) AND gate is _____.
a) A/B b) A-B
c) A+B d) A.B

2) A J-K flip-flop made to toggle _____.
a) J = 0, K = 0 b) J = 1, K = 0
c) J = 0, K = 1 d) J = 1, K = 1

3) _____ are universal gates.
a) NOT b) NAND & NOR
c) AND d) NOT, AND, & OR

4) 8085 is _____ bit microprocessor.
a) 8 bit b) 16 bit
c) 32 bit d) 4 bit

5) _____ is branch control group of instruction.
a) MOV b) ADD
c) CALL d) RAR

6) Address bus of 8085 is _____ bit.
a) 8 bit b) 16 bit
c) 32 bit d) 4 bit

Q.2 Answer the following. (Any Three) **06**

a) Define logic gates.
b) Explain D flip-flop.
c) Draw symbol of AND, OR, NOT and EX-OR.
d) List data transfer group of instructions.
e) Define Machine cycle and instruction cycle.

Q.3 Answer the following. (Any Two) **06**

a) Write features of 8085.
b) Explain Half Adder.
c) Explain 4:1 multiplexer.

Q.4 Answer the following. (Any Two) **06**

a) Explain system bus architecture of 8085.
b) Write a program for 8 bit Addition.
c) Explain RS flip-flop.

Q.5 Answer the following. (Any One) **06**

a) Explain Pin diagram of 8085.
b) Explain Full adder with suitable diagram.